Determining Satellite Reliability using the Kaplan Meier Estimator and the Cox Proportional Hazard Model

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Contents

1 Introduction and Motivation

Satellites have played an important role in sustainability and understanding the state of the world. Given the ability to offer fact-based information and imagery, satellites can solve some of Earth's greatest challenges like climate change [**?**]. There are many examples of how Satellites have been used to develop and sustain life on earth such as monitoring reservoir water and identifying crop yields across a vast territory. The engineering and development of a satellite system heavily depend on the objective of the mission. Careful consideration of each step of the design process is important for the success of the mission. Therefore, it is critical for satellite engineers to consider the reliability of a satellite and its subsystems.
Reliability is concerned with the time of correct performance (non-failure) of a system under specific conditions to ensure that a system will perform its intended task [1]. Reliability analysis has been a critical component in various parts of space system development, for example, tests are conducted before launch to identify and eliminate any possible causes for on-orbit failures [5]. furthermore,

Engineers are not only interested in the correct performance of a system, but also the time until the occurrence of a failure. Typical Satellite data consists of launch dates, and failure times, where the event of interest is a failure. If the event of interest is not observed for a particular observation, it is known as censoring. These features of the data make survival analysis the chosen method for analysing satellite data [4] Survival analysis includes the use of statistical methods for analysing the time to the occurrence of an event of interest. [12].

There are several methods used to analyse survival data but for this research, the focus will be on the Kaplan Meier estimator and the Cox proportion hazard model. The Kaplan Meier method is the simplest survival method often used to analyse time-to-event of interest. This method is based on conditional probability and involves computing the probability of the event occurring (i.e. Failure of a satellite), at a certain point in time, and multiplying that probability with the probability of surviving the previous time interval [8]. The limitation with the use of the Kaplan Meier model is that it does not take into account effect variables (confounders or covariates) which can potentially contribute to failure. However, The Cox proportional hazard model is often used to address scenarios where the presence of specific risk factors can affect the survival of a system [16]. The Cox proportional hazard model is a linear model used to model the hazard (failure rate) given the presence of explanatory variables. furthermore, The CPH model investigates the association between failure rate and explanatory variables, and how these factors measure and explain differences between sample groups [6].

1.1 Aims and Objectives

According to [5] "no two (or more) satellites are truly alike", for example, satellites come in different sizes, and operate in different locations. As a result, analysing the collective reliability may not accurately reflect the reliability of a specific spacecraft. Therefore, this research aims to analyse the reliability of satellites by categorising satellites according to size (mass in kg) and orbit location. The objective of this study is to analyse the reliability behaviour of satellites in a particular orbit, given the size of the satellite. This study attempts to answer the following questions:

Question 1: Do satellites of different sizes (small, medium, and large) operating in Low-earth orbit exhibit the same reliability profile?. This question is also extended to Geosynchronous and medium-earth orbiting environments.

Question 2: Do small satellites operating in different orbit environments exhibit different reliability profiles?. This question can be extended to medium and large satellites.

2 Literature Review

Despite the importance of satellite reliability, there have not been a lot of statistical analysis and satellite reliability studies conducted in the literature. The main reason for this is due to limited on-orbit failure data. However, authors [9] managed to fill this gap by conducting a study dedicated to the statistical reliability of small satellites. The study made use of non-parametric and parametric approaches. Given the highly censored data, the non-parametric method used was the Kaplan Estimator. To analyse the failure data effectively, the authors made use of the Weibull distribution to model the reliability and use the model for optimisation and forecasting purposes. To estimate the Weibull parameters, two approaches were used and compared, namely the Maximum

Likelihood Estimation and a combination of the Bayesian Theory with Markov Chain Monte Carlo simulations. The results of the study show that the scale parameters estimated using Bayesian methods were much lower than the scale parameter estimated using MLE. Furthermore, the reverse was true for the shape parameter, where estimations obtained using Bayesian methods were higher than those obtained using MLE. In addition to this, the coefficients estimated using the Bayesian approach resulted in a better-fitted Weibull model. Therefore, The Bayesian approach with the Markov chain approach shows great potential for satellite reliability.

The approaches mentioned above have been used in various other fields of study such as clinical treatment trials and biology. In the following study, [15] the Kaplan Meier method was used to examine mortality in patients starting renal replacement theory. In this study, the Kaplan Meier is primarily used to compare the mortality of patients of two distinct groups. The resulting cumulative survival probabilities calculated using the KM method were then plotted on the same graph to examine and compare the survival curves. The graphical results show differences in the survival profiles of the two groups. To further examine if there is a difference between the two groups, the log-rank test was used to test the statistical significance. Lastly, the study highlights the limitations of using the KM method and the log-rank test being that you cannot assess the size of the difference between the groups and the KM method does not offer for the adjustment of other factors which will allow for a fair comparison between the two groups. The KM method is an effective method to test and present the differences between population groups, however, the disadvantage arises in not being able to quantify this difference and test the effect of different variables [15].

Another popular approach that is commonly used in contrast to the Kaplan Meier method is the Cox regression method. In the following clinical study, [15] the Cox regression model is used to examine the association between eGFR at the start of dialysis and all-cause mortality in patients partaking in dialysis. The study builds 4 cox regression models to compare the mortality risk of two or more groups. For example, the first model categorises patients according to high, medium, and low eGFR groups, and model 2 categorizes and compares patients in high-medium and low eGFR groups. In each cox regression model, the hazard rates are determined and a hazard ratio is derived to estimate the survival probabilities. furthermore, the Cox regression model is also used for components of mechatronic systems and components such as in the study conducted by [2] where the research objective is to transfer predictions for new components from existing datasets, where the new components were built with the same technology as existing components. The study fits a cox regression model with data from in-house experiments and Monte Carlo simulated data. To retrieve the reliability data, the researchers performed endurance tests where failure data was observed. The results of the study show that the cox model is effective in reliability prediction. Moreover, the authors recommend the careful consideration of covariates, to ensure that covariates are valid for more than one sample group.

In the following study, [**?**] two models were used to estimate the effect of risk factors on Tuberculosis survivors. The study uses a logistic model and a Cox regression model in which its hazard function was estimated using the Kaplan Meier estimator. Two model evaluation techniques were used namely the Akaike Information Criterion and Bayesian Information Criterion. The results of the study show that the logistic model is the best model to use, and the Cox regression was effective in identifying the most common factors that have an impact on tuberculosis. The Kaplan Meier and Cox regression have been mostly used in clinical trials and biology. Therefore, investigating satellite reliability using these models has the potential to offer a different perspective of how the models can be used.

3 Methodology

The purpose of this study is to investigate the effect of satellite mass and orbit environment on the reliability of satellites. The objective is to use the two most commonly used survival analysis techniques namely the Kaplan-Meier method and the Cox regression approach to investigate this problem. Furthermore, these two methods will be compared to determine which model best predicts reliability and captures the survival profile of satellites.

Data Categorisation

There are many variables that contribute to the failure behaviour of a satellite but for the purpose of this study, only two variables will be investigated;

- **Satellite size according to mass in Kg** The following categorisation of satellite mass is taken from [5]:
  - Small- 0 to 500 kg
  - Medium- 500 to 2500 kg
  - Large- 2500 <
- **Orbit environment**
  - GEO-Geosynchronous orbit (35 000km above Earths surface)
  - LEO-low earth orbit (2 000 km above Earth's surface)
  - MEO-medium Earth orbit (10 000 to 20 000 km above Earths)

To effectively answer the research questions and to investigate the different reliability behaviours, the data must be split into categories.

The following categories will be used to investigate Question 1:

Category 1.1: GEO Orbiting satellites

This category includes small, medium, and large satellites operating in geosynchronous earth orbit

Category 1.2: LEO orbiting Satellites

This category includes small, medium, and large satellites operating in low-earth orbit **Category 1.3**: MEO orbiting satellites

This category includes small, medium, and large satellites operating in medium earth orbit

The following categories will be used to investigate Question 2;

Category 2.1: Small satellites This category includes small satellites operating in different orbit environments

Category 2.2.: Medium satellites This category includes medium satellites operating in different orbit environments

Category 2.3.: Large Satellites This category includes large satellites operating in different orbit environments

3.1 Censoring

Survival analysis involves the analysis of data between two specified intervals such as birth and death. For this study, the interval of interest is between successful orbit insertion and failure of the satellite. The nature of survival data is different from data in other areas of statistics in that it consists of censored or incomplete data[13]. In the context of survival analysis, censoring occurs when information about the event of interest is not available or the event of interest is not observed for that particular item[7]. Different situations can result in the occurrence of censored data.There are different types of censoring mechanisms such as point censoring and interval censoring. For this research, the focus will be on point censoring which refers to situations where survival time and censoring time are known [13]. For this study Satellite data is cencored for two reasons, first, satellites that have completed their missions during the observation window are retired before the failure time. This is known as right censored data, as the event of interest will not be observed. In addition, censoring may also occur when the satellite is still operational at the end of the observation window. In both these cases, failure will not be observed, however, survival time and censoring time are known [10]. When there is presence of incomplete data it is important to find the right statistical methods that can handle censored data. For this research The Kaplan Estimator is used as the non-parametric estimation method to handle the type of censoring mentioned. [15]

3.2 Kaplan Meier Method

The Kaplan Meier estimator is a non-parametric estimator used to analyse time-to-event. This method is built on three assumptions [8];

- Assumption 1:At any time observations who are censored have the same probabilities as those who are not censored
- Assumption 2:Survival probabilities remain the same for all observations despite the start date.i.e. observations that are recruited early and late have the same survival probabilities.
- Assumption 3: Recorded events happen at the exact time it is specified

The concept behind the KM estimator is conditional probability and can be interpreted as the probability of surviving a specific time interval given the survival of previous time intervals [15]; this concept can be expressed in the following manner taken from [8]:

$$S_i = \frac{\text{number of subjects living at the start - number of subjects that have failed}}{\text{number of subjects living at the start}}$$

The above expression states that the survival probability S_i at each time interval is calculated as the number of subjects operating divided by the number of subjects operating from the start.
based on this understanding, the Kaplan estimator without censored data is mathematically expressed as:

$$\hat{R}[t] = 1 - \frac{i}{n} \text{ [1]}$$

where $i = 1, 2, ..., n$ is the number of failed satellites at time t
For example, suppose at time t=2 years, 1/23 satellites failed. Thus the survival probability at t=2 is calculated as:

$$\hat{R}(2) = 1 - \frac{1}{23}$$
$$\hat{R}(2) = 1 - 0.0434$$
$$\hat{R}(2) = 0.9566$$

The above expression producers the survival probability without censored data. When censored data is considered, the Kaplan Estimator is expressed as [14];

$$\hat{R}(t) = \prod_{i=1} \frac{n_i - 1}{n_i}$$
[2]

- n_i denotes the number of operational unites right before t_i
- $n =$ the number of censored units right before t_i - the number of failed units right before t_i

However, suppose censoring time is exactly equal to failure time, then censoring is assumed to occur immediately after the observed failure time. In this case, the Kaplan estimator can be expressed as follows [5];

$$\hat{R}(t) = \prod_{i=1} 1 - \frac{m_i}{n_i}$$
[3]

- m_i denotes all the units failing at exactly t_i

Now that the KM estimator is defined, it can be applied to the data.

Table 1 Showns an example of how the application of the KM estimator.(note: arbitrary values were used)

- Column 1-Time of the event(failure time)
- Column 2-Number of failed satellites in time t_i
- Column 3-Number of satellites at risk in time t_i. Satellites at risks refer to all the satellites at the start of t_i at risk of failure but have not failed
- Column 4-Number of censored satellites at t_i
- Column 5-Shows the probability of surviving t_i
- Column 6-shows the cumulative probability

Table 1: Kaplan-Meier estimate for small satellites in GEO

Time of Event	**no. of failed satellites**	**no. of satellites at risk**	**no. of censored satellites**	**Probability of surviving t_i**	**cummulative probability**
0	0	50	0	$\frac{50}{50} = 1$	1
3	1	50	0	$\frac{50-1}{50} = 0.98$	$1x0.98 = 0.98$
4	1	49	0	$\frac{48}{49} = 0.9796$	$0.98x0.9796 = 0.96$
8	0	47	1	$\frac{47}{47} = 1$	$0.96x1 = 0.96$

For the purpose of this research, table 1 will be reconstructed to produce tables for the following categories:

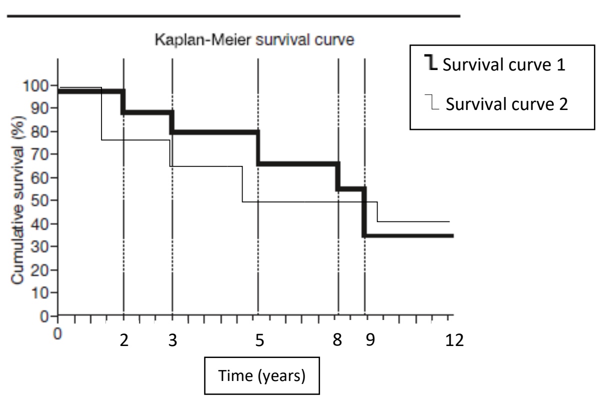
- Table 1.2: The KM estimator for small satellites in LEO
- Table 1.3: The KM estimator for small satellites in MEO
- Table 1.4: The KM estimator for small satellites in GEO
- Table 2.-2.3: The KM estimator for medium satellites in GEO, LEO and MEO
- Table 3.1.-3.3:The KM estimator for large satellites in GEO, LEO and MEO

In total there will be 9 tables with KM estimators for different categories.

3.3 Kaplan Meier Survival curves

One of the objectives of this study is to investigate the survival profiles of satellites in different orbital environments given their size in mass. Therefore using the cumulative survival probabilities calculated using the KM method, the survival curves for each category will be constructed. On the x-axis, the time in years will be presented and on the y-axis, the cumulative probabilities taken from the last column of each table will be presented. To plot the different curves, two coordinates are needed. The first coordinate represents the time in which the event of interest occurred t_i. The second coordinate represents the cumulative probability at t_i. The resulting curve will have steps that demonstrate the decrease in cumulative survival whenever an event occurs.
Figure 1 shows an example of two survival curves plotted on one graph. The focus in this study is to compare 3 survival curves simultaneously, for example, the survival curve of small satellites in GEO, MEO, and LEO. Therefore, in total there will be six KM survival plots, each comparing three survival curves;

Figure 1: Example of Kaplan-Meier survival curves



- Figure 1.1: Comparing survival curves of small satellites in GEO, LEO, and MEO
- Figure 1.2: Comparing survival curves of medium satellites in GEO, LEO, and MEO
- Figure 1.3: Comparing survival curves of large satellites in GEO, LEO, and MEO
- Figure 2.1: Comparing small, medium, and large satellites in GEO
- Figure 2.2:Comparing small, medium and large satellites in LEO
- Figure 2.3: comparing small, medium, and large satellites in MEO

The Kaplan Meier plots offer a visual representation of reliability and orbit failure, and thus by plotting the different survival curves simultaneously, an investigation of the differences can be conducted. To test whether the survival curves are statistically different a log-rank test can be used to compare survival curves of two or more independent groups [15]. The Log-rank test is a hypothesis test testing the null hypothesis of no differences between two or more observed groups. The limitations of the following method are that it tests the survival at one point in time, and does not consider the entire observation window [11]. Furthermore, both the KM plots and log-rank test do not offer information on the size of the differences between the groups. Therefore, The KM method is effective in plotting survival curves and visualising differences between groups, however, it does not account for explanatory variables.

3.4 Cox Proportional Hazards model

The purpose of this study is to determine the effect of satellite mass and orbit environment as factors contributing to on-orbit failure, and thus determine how these factors affect satellite reliability. Therefore, to further investigate these predictors, it is of interest to make use of a model which takes into account effect variables such as the Cox Proportional hazards model[**?**]. The Cox proportional hazard model is a regression model used to investigate the relationship between the time-to-event and one or more explanatory variables[14]. What distinguishes the CPM from other survival models is that it takes into account the risk factors/predictors/covariates associated with the event of interest. The explanatory variables or covariates are variables that explain the object under consideration and environmental conditions. This means, for each i there are k corresponding covariates [2]. The CPH model models the relationship between the hazard rate and covariates in the follow way [6];

$$H(t) = H_0(t) + \exp(\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + ... + \beta_k x_k) \text{ [1]}$$

where:

- $x_1, x_2, ..., x_k$ are the predictors
- $H_0(t)$ is the baseline hazard of an individual or sample group being considered. The baseline hazard is the hazard present when all $x_i = 0$

In the context of satellite failure, the predictors chosen to explain the hazard rate are:

- size of the satellite

- orbit location

Therefore equation [1] can be re-written to define the hazard rate of satellites as [16];

$$H(t) = H_0(t) + \exp(\beta_1 x_1 + \beta_2 x_2)$$

where:

- x_1 takes in the size of the satellite
- x_2 takes orbit location

To apply the CPH model, the β_i coefficients need to be estimated. Generally, the $\hat{\beta}_i$ estimates under the CPH model are estimated using the maximum likelihood approach together with the Newton-Raphson technique used for maximization [14]. Once the $\hat{\beta}_i$ have been estimated, the hazard ratio and corresponding survival functions can be derived and estimated. The hazard rate can be expressed in matrix notation as [17];

$$h(t, X) = h_0(t) + \exp \sum(\beta_i X_i)$$

where:

- $h(t, X)$ denotes the hazard function
- β_i denotes the regression coefficients
- X_i represents the matrix of covariates used to model the failure rate

One of the most important calculations derived from the CPH model is the hazard ratio (HR). The hazard ratio produces a real-valued number that describes the relationship between two groups concerning their hazard functions over time. To calculate the HR, the following equation can be used;

$$\hat{HR} = \frac{\hat{h}(t,X)}{\hat{h}(t,X*)}$$

where:

- $\hat{HR}$ denotes the estimated hazard ratio
- $\hat{h}(t, X)$ denotes the hazard rate for group 1
- $\hat{h}(t, X^*)$ denotes the hazard rate for group 2
- X and X^* denotes the set of risk factors or covariates for each group respectfully.

The resulting $\hat{HR}$ can be interpreted as follows;

- $\hat{HR} > 1$ This means the hazard rate of group 1 is higher than the hazard rate of group 2. i.e. group 1 experiences a higher failure rate than group 2
- $\hat{HR} < 1$ This means the hazard rate of group 2 is higher, i.e. group 2 experiences higher failure rate than group 1
- $\hat{HR} = 1$ This means both groups experience the same failure rate overtime

Numerically, if the $\hat{HR} = 1.3$ This means that group 1 (e.g.satellites operating in low-earth orbits) experience a 30% higher failure rate than group 2 (satellites operating in geosynchronous orbit). Therefore, unlike the log-rank test, the Hazard ratio can quantify the difference between two groups. To further investigate the reliability profiles of the different satellite groups, the hazard function can be used to derive the survival probabilities. This is expressed mathematically as [14];

$$\hat{S}(t, X) = \hat{S}_0(t) + \exp \sum(\beta_i X_i)$$

where:

- $\hat{S}(t, X)$ denotes the survival at time t_i adjusted for the explanatory variables
- $\hat{S}_0$ denotes the baseline survival function

One assumption of the CPH model is that the hazards of two groups being compared should always be proportional to each other over time and if this assumption is violated it may lead to inaccurate estimates[15]. One way to visually test this assumption is by plotting the log-log survival functions of the two groups in question [14]. If the survival curves of the two groups overlap then the assumption has been violated and the use of the CPH model is not appropriate in this scenario [15].

4 Model Evalulation

In the previous section, two models were defined to estimate survival probabilities, the Kaplan Meier estimate, and the Cox proportional model. Suppose in the testing phase of satellite design spacecraft engineers want to know the probability of failure of a small satellite operating in LEO vs a medium satellite operating in LEO after 5 years in orbit. Keeping this in mind, it is important that the chosen model is the best fit for the data and the most accurate for prediction. Thus, model evaluation is an important step that should be conducted before using a prediction model. Several methods can be used to evaluate the prediction accuracy of survival models such as Akaike's Information Criterion, generalised R^2 and the Bayesian Information Criterion [**?**]. For this study, the focus will be on Akaike's Information Criterion.

4.1 Akaike's information Criterion

Akaike's Information Criterion or AIC is a model selection criterion used to compare the quality of models to select the model that best fits the data. The AIC looks at the different variables used in the model and the maximum likelihood estimate [**?**]. Generally the AIC is mathematically expressed as;

$$AIC = -2ln(\hat{L}) + 2K$$

Where:

- K= the number of parameters in the model
- $ln\hat{L}$log- is the likelihood measure of model fit

However, For the purpose of this study, censored data needs to be accounted for. Therefore, an improved version of the AIC must be considered. According to [**?**], the following likelihood function can be considered for the improved AIC;

$$L = \prod_u h(Z_i|x_i) \prod_{i=1}^{n} S(Z_i|x_i)$$

where:

- n is the total number of observations
- T, C and x denotes the survival time,censoring time
- $Z = min(T, C)$ is the observed time
- $h(t|x) and S(t|x)$ be the conditional hazard and survival functions

Therefore, the improved AIC is expressed as;

$$AIC = -2ln(\hat{L}) + 2(p + 2 + K)$$

where;

- p denotes the number of covariates
- k denotes the number of parameters

Once both models have been fitted and the corresponding parameters have been estimated, the AIC values can be computed using the above equations. The response variable T for this study is satellite failure time and the chosen covariates used are;

- X_1-Satellite size (mass in kg) (reference- small,1-medium, 2-large)
- X_2-Orbit Location (reference-LEO, 1-MEO, 2-GEO)

When X_1,X_2 = 0 this denotes a small satellite operating in low-earth orbit. According to [**?**], the best fitting model is the one with the smallest AIC score, which translates to the model that explains the most variations with the least amount of parameters.

5 Conclusion

In conclusion, to make a fair comparison between the Kaplan Meier estimator and the CPH model, both models need to be parametric and take into account explanatory variables. In the current study the Kaplan Meier estimator is non-parametric and can only take on one variable at a time, this may pose a challenge when evaluating the two models. A possible solution is to fit a single or 2-Weibull mixture model using the KM estimates. Therefore, transforming the KM estimator to a Weibull distribution may offer better opportunities for optimisation and forecasting [3]

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